

ALP

WORKSHEET
TEACHING ACTIVITY

Algorithmics and Data Structures

COURSE UNIT

worksheet 07 – File handling

APPLICATION 1 – TEXT FILE

Create an application that allows you to manage a country file (***países.txt***), with a structure like the one shown in the example: *country; continent*

Your app should include a menu with the following options:

C:\Users\mario\OneDrive\AED\4 - Exercicios\Ficha 1

```
MENU
1 - Inserir Países
2 - Consulta Países
3 - Consulta por continente
4 - Consulta nº países
0 - Sair

opção: _
```



1- Inserir Países | Add Countries

Call a function names ***AddContry()*** that:

- It asks the user to enter a country and a continent;
- If the ***países.txt*** file does not exist, your application must create it, in a sub-folder ***files*** within your project folder.
- Check to see if the country already exists in the file. If there is, you must show a message to the user and not duplicate the country name in the file.

- If the country does not exist in the file, you must add it to the country file.

c:\Users\mario\OneDrive\AED\4 - Exercicios\Ficha 09 - VS Code Console

```
País      : Portugal
Continente : Europa_
```

2- Consulta Países | View Countries

- You must call a function ***showCountries()*** that allows you to list all the countries in the países.txt file.
- If your file has more than 10 countries, please list 10 countries per page.

```
C:\WINDOWS\py.exe

País      Continente
-----
Portugal  Europa
Espanha   Europa
Suécia    Europa
Finlandia Europa
Brasil    América
Japão     Asia
China     Asia
Peru      América
Guatemala América
Grecia    Europa

Pág. 1. Prima <enter> para continuar
```

3- Consulta por continente | By Continent

- You must ask the user to indicate the continent.
- Next, you should call a function (which receives the continent indicated by the user) that allows the countries of that continent to be listed on file.

c:\Users\mario\OneDrive\AED\4 - Exercicios\Ficha 09 - VS Code

Continente: Europa_

c:\Users\mario\OneDrive\AED\4 - Exercicios\Ficha 09 - VS Code Console

País	Continente
Portugal	Europa
Espanha	Europa
França	Europa
Itália	Europa
Suécia	Europa

4- Consulta nº Países por Continente | Number of Countries by Continent

- This option should call a function that prints the number of countries, on file, by *each continent

APPLICATION 2: BINARY FILE

1. Implement the **writeText(text)** function that receives a text (read, e.g. through an input) and saves it in a binary file, called *text.bin*. If the file does not exist, it must be created in a subfolder *files*.
2. Implement the **readText()** function that reads the contents of the *./files/text.bin* file and returns the corresponding text. Print the result, returned by the function, on the console application.

Verify that the file exists and that no errors occur during its reading.

Note: the *path* and filename should be easily parameterized in your code (global variables), without the need to change the code of the functions requested above.